DR. A.P.J ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW



EVALUATION SCHEME & SYLLABUS

FOR

B. TECH. III YEAR

(ENVIRONMENTAL ENGINEERING)

On

Choice Based Credit System

[Effective from session 2018-19]

FIFTH SEMESTER ENVIRONMENTAL ENGINEERING SESSION 2018-19

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					ESE	CT	TA.		
1	RAS501	MANEGIRIAL ECONOMICS	Applied Science	3—0---0	70	20	10	100	3
2	RAS502/ RUC501	SOCIOLOGY /CYBER SECURITY	Applied Science	3—0---0	70	20	10	100	3
3	REV 501	AIR AND NOISE POLLUTION CONTROL ENGINEERING	Core Deptt.	3—0---0	70	20	10	100	3
4	REV 502	DESIGN OF CONCRETE STRUCTURE	Core Deptt.	3—1---0	70	20	10	100	4
5	REV 503	SOLID WASTE MANAGEMENT	Core Deptt.	3—0---0	70	20	10	100	3
6	RCE 051 RCE052 RCE 053	**ELECTIVE -1** MODERN CONSTRUCTION MATERIALS CONCRETE TECHNOLOGY GEOENVIRONMENTAL ENGINEERING	Core Deptt.	3—1--0	70	20	10	100	4
7	REV 551	AIR POLLUTION CONTROL ENGINEERING LAB	Core Deptt.	0—0---2	50		50	100	1
8	REV 552	ENVIRONMENTAL SANITATION LAB	Core Deptt.	0—0---2	50		50	100	1
9	REV 553	SOLID WASTE MANAGEMENT LAB	Core Deptt.	0—0---2	50		50	100	1
10	REV 554	BUILDING PLANNING AND DRAWING LAB	Core Deptt.	0—0---2	50		50	100	1
	TOTAL				620	120	260	**1000**	**24**

REV 501: Air and Noise Pollution and Control Engineering (L-T-P-3-0-0) Credit - 03

Unit I:
Air pollution: composition and structure of atmosphere, global implications of air pollution. Classification of air pollutants: particulates, hydrocarbon, carbon monoxide, oxides of sulphur, oxides of nitrogen and photo chemical oxidants. Indoor air pollution, Effects of air pollutants on humans, animals, property and plants. **[8]**

Unit II:
Air pollution chemistry, meteorological aspects of air pollution dispersion; temperature lapse rate and stability, wind velocity and turbulence, plume behaviour, dispersion of air pollutants, the Gaussian Plume Model, stack height and dispersion. **[8]**

Unit III:
Ambient air quality and standards, air sampling and measurements; Ambient air sampling, collection of gaseous air pollutants, collection of particulate air pollutants, stack sampling. Control devices for particulate contaminants: gravitational settling chambers, yclonee separators, wet collectors, fabric filters (Bag-house filter), electrostatic precipitators (ESP). **[8]**

Unit IV:
Control of gaseous contaminants: Absorption, Adsorption, Condensation and Combustion, Control of sulphur oxides, nitrogen oxides, carbon monoxide, and hydro carbons. Automotive emission control, catalytic convertor, Euro-I, Euro-II and Euro-III specifications, Indian specifications. **[8]**

Unit V
NOISE POLLUTION: Basics of acoustics and specification of sound; sound power, sound intensity and sound pressure levels; plane, point and line sources, multiple sources; outdoor and indoor noise propagation; psycho-acoustics and noise criteria, effects of noise on health, annoyance rating schemes; special noise environments: Infra-sound, ultrasound, impulsive sound and sonic boom; noise standards and limit values; noise instrumentation and monitoring procedure. Noise indices. **[8]**

References:
1. Peavy, Rowe and Tchobanoglous: Environmental Engineering.
2. Martin Crawford: Air Pollution Control Theory.
3. Warkand Warner: Air Pollution: Its Origin and Control.
4. Rao and Rao: Air Pollution Control Engineering.
5. Keshav Kant and Rajni Kant, "Air Pollution and Control Engineering", Khanna Publishing House.
6. Environmental Pollution Control Engineering-CS Rao, Wiley Eastern Ltd., New Delhi, 1996.
7. C.S. Rao, Air pollutionand control
8. Environmental Noise Pollution – PE Cunniff, McGraw Hill, New York, 1987
9. Nevers: Air Pollution Control Engineering.
10. M. P. Poonia and S C Sharma," Environmental Engineering, Khanna Publishing House.

11. My cock, Mc Kenna and Theodore: Handbook of Air Pollution Control Engineering and Technology. Suess and Crax ford: W.H.O. Manualon Urban Air Quality Management
10. OP Gupta,Elements of Environmental Polluton Control, Khanna Publishing House.

REV 502: Design of Concrete Structures (L-T-P 3-1- 0) Credit - 04

Unit 1

Concrete technology: Constituent materials and their Properties, Fresh concrete, Work ability, strength, elasticity and fracture of hardened concrete, Time dependent properties of concrete, Durability of concrete, concrete and mixtures, Mix design methods, BIS code provisions, Special concretes.

Introduction to various design philosophies, Assumptions in working stress and limit state design method, Factor of safety applied to different materials, IS-456 provisions, Load consideration for design of concrete structures as per relevant IS code. **[8]**

Unit 2

Beam design by limit state: Moment of resistance of a beam of rectangular and flange section and their design using limit state, Spacing of vertical and inclined stirrups, Development length of reinforcement, Detailing of reinforcement as per IS-456 and IS-13920. **[8]**

Unit 3

Column design by limit state: Short column under axial loading and uni-axial bending, column with epical reinforcement. Detailing of reinforcement as per IS-456 and IS-13920. Solid Slab design using limit state: Design of one way and two way slab. **[8]**

Unit 4

Retaining wall: Structural behaviour, stability of retaining wall against over turning and sliding, Design of T-shaped retaining wall, Concept of counter for retaining wall. **[8]**

Unit 5

Water Tanks: Design of circular and rectangular tanks situated on the ground, underground and over head tanks. **[8]**

Note: The students should be given a comprehensive problem at the end which requires inputs/ knowledge/ application from all the units of the syllabus. It may be evaluated as a part of TA.

Books

1. Jain ,O.P. & Krishna, J."Plain and Reinforced Concrete, Vol. I & II", Nem Chand & Bros.
2. Dayaratnam, P. "Reinforced Concrete Design", Oxford & IBH.
3. Sinha, S.N. "Reinforced Concrete Design'', Mc- Graw Hill.
4. Pillai, S.U. & Menon, D. "Reinforced Concrete Design, Mc- Graw Hill.
5. Gambhir, M.L. "Fundamentals of Reinforced Concrete Design", PHI Learning.

6. Park ,R. and Pauley, T. "Reinforced Concrete Structures", John Wiley.
7. IS: 456–2000: Code of Practice for Plain and Reinforced Concrete, Bureau of Indian Standards.

REV 503: Solid Waste Management `(L-T-P 3-0-0) Credit- 03

Unit-1

Sources and types of solid wastes, material flow and waste generation, Functional elements: Waste generation, storage, collection, Transfer and transport, processing and recovery, disposal. Physical and chemical composition of municipal solid waste integrated solid waste management, hierarchy of waste management options. **[8]**

Unit-2

Different methods for generation rates. **Storage:** movable bins, fixed bins. **Collection:** home to home collection, community bin system. Theory and design of hauled container system, stationary container system **Transportation:** handcart, tri-cycle, animal cart, tripper truck Transportation, dumper placer, bulk refuse carrier, railroad transport, Water transport, conveyors, layout of routes. **[8]**

Unit-3

Engineering system for on-site **handling and processing** of solid waste: separators size reduction equipments, screening equipments densification, baling, cubing, peleting equipments. **[8]**

Unit-4

Land filling: Site selection criteria, landfill layout, landfill sections, Occurrence of gases and leachate in landfills: composition and characteristics, generation factors of landfill gases and leachate, initial adjustment phase, transition phase, acid formation phase, methane formation phase, maturation phase of gases, initial adjustment phase, transition phase, acid formation phase, methane formation phase, maturation phase of leachate. **[8]**

Unit-5

Composting, types of composting, process description, design and operational consideration of aerobic composting, process description, design and operational consideration of anaerobic composting, Thermal conversion technologies: incineration and pyrolysis system, energy recovery system, Electronic waste and Biomedical waste Overview of solid waste management practices in India. **[8]**

Reference books:

1. Tchobanoglous, G., Theisen, H., & Vigil, S.A; Integrated Solid Waste Management: McGraw Hill, New York
2. Solid Waste Engineering, Principle & Management issues by VenTe Chow
3. Bhide, A.D., B.B. Sundaresan, Solid Waste Management in developing countries.
4. Manual on Municipal solid Waste Management, CPHEEO, Govt. of India.
5. Guidelines for Management and Handling of Hazardous wastes MOEF (1991), Govt. of India.
6. Datta, M; Waste Disposal in Engineered Land fills, Narosa Publishers, Delhi.

7. OP Gupta, Elements of Solid Hazardous Waste and Management, Khanna Publishing House.
8. Waste Management "Asian and Pacific Center for Transfer of Technology (N.D.) India", September 1993.

RCE 011 Modern Construction Materials (L-T-P 3-1-0) Credit - 4

Unit – 1

Introduction, properties and uses of modern building materials: fly ash bricks, soil - cement blocks, calcium silicate bricks, red mud jute fibre polymer composite (RFPC) , glass reinforced gypsum. **[8]**

Unit – 2

Introduction , properties and use of: geosynthetics, bituminous material, fire resistant materials (chemicals ,paints ,tiles ,bricks, glass),metals, light - weight concrete, mass concrete, waste material based concrete. [**8]**

Unit – 3

Introduction , properties and use of: Ferro cement & fibre reinforced concrete, different types of fibres, high density concrete, Nuclear concrete, heat resisting & refractory concretes, pre fabricated systems. **[8]**

Unit – 4

Introduction, properties and use of: Polymers, fibre reinforced polymers, polymer concrete composites (PCCs), sulphur concrete and sulphur - infiltrated concrete. **[8]**

Unit – 5

Introduction, properties and use of: Conventional and modern water proofing materials, Conventional and modern insulating materials (thermal, sound and electrical insulating materials).Concept of polymer floor finishes. **[8]**

Reference Book:

1) GhambhirM.L."Concrete Technology" Tata McGraw Hill education private Limited.
2) A.R. Santhakumar,Concrete Technology,Oxford University Press ,2006.
3) Building Materials, P.C. Varghese, Prentice-Hall India, 2555.
4) Shetty,M.S.,"Concrete Technology" S. Chand Publication.
5) Krishnaraju .N., Advanced Concrete Technology,CBS Published,1985.
6) Materials Science and Engineering: An introduction, W.D. Callister, John Wiley, 1994.
7) Nevile. A.M., Concrete Technology, Prentice Hall,Newyork, 1985.
8) Dr. U. K. Shrivastava, Building Materials Technology, Galgotia Publication pvt.ltd.

9) Materials Science and Engineering, V. Raghavan, Prentice Hall, 1990.
10) Properties of Engineering Materials, R.A. Higgins, Industrial Press, 1994.
11) Construction materials: Their nature and behaviour, Eds. J.M. Illston and P.L.J. Domone, 3rd ed., Spon Press, 2551.
12) The Science and Technology of Civil Engineering Materials, J.F. Young, S. Mindess, R.J. Gray& A. Bentur, Prentice Hall, 1998.
13) Engineering Materials 1: An introduction to their properties & applications, M.F. Ashby and D.R.H. Jones, Butterworth Heinemann, 2553.
14) The Science and Design of Engineering Materials, J.P. Schaffer, A. Saxena, S.D. Antolovich, T.H. Sanders and S.B. Warner, Irwin, 1995.
15) Concrete: Microstructure, properties and materials, P.K. Mehta and P.J.M. Monteiro, McGraw Hill, 2556.
16) SK Sharma, "Civil Engineering construction material" Khanna Publishing House.
17) Properties of concrete, A.M. Neville, Pearson, 2554.

RCE 012 Concrete Technology (L-T-P 3-1-0) Credit - 4

Unit I

Cement: production, composition properties, types and cement chemistry. Introduction to supplementary cementitious materials. Aggregates: mineralogy, properties, test and standards. Quality of water for use in concrete. **[8]**

Unit II

Introduction & study of accelerators, retarders, water reducers, air entrainers , water proofers, super plasticizers. Study of supplementary cementing materials like fly ash, silica fume , ground granulated blast furnace slag, metakaoline and pozzolana; their production, properties and effect on concrete properties . **[8]**

Unit III

Principle of mix proportioning, properties related to mix design, Mix design method (IS method and ACI method). Mix design of concrete: packing density, Rheology, mix design examples. **[8]**

Unit IV

Concert production, batching, mixing and transportation of concrete. Workability: test for work ability of concrete (slump test, compacting factor test and Vee Bee test). Segregation and bleeding in concrete, curing of concrete and its methods. Determination of compressive and flexural strength as per BIS. Mechanical properties of concrete: elastic modules, poisson's ratio, creep, shrinkage and durability of concrete . **[8]**

Unit V

Study and uses of high strength concrete, self compacting concrete, fiber reinforced concrete, ferro cement, ready Mix Concrete, recycled aggregate concrete and status in India. **[8]**

References

1. Neville, A.M. and Brooks, J.J.," CONCRETE TECHNOLOGY", ELBS .1990.
2. Shetty, M.S, "Concrete Technology, Theory and Practice", S. Chand and Company Ltd, New Delhi, 2008.
3. Gambhir, M.L, "Concrete Technology", Tata McGraw Hill Publishing Company Ltd, New Delhi, 2004.
4. Santha kumar, A.R; "Concrete Technology", Oxford University Press, New Delhi, 2007.
5. Gupta B.L., Amit Gupta, "Concrete Technology", Jain Book Agency, 2010.
6. Newman, K., "CONCRETE SYSTEMS in COMPOSITE MATERIALS".EDT BY L. Holliday. Elsevier Publishing Company. 1966.
7. Popovics.S., "FUNDAMENTALS OF PORTLAND CEMENT CONCRETE: A Quantitative Approach VOL 1 FRESH CONCRETE" JOHN WILEY & SONS.1982.
8. P.K. Mehta and Paulo J.M. Monteiro, "Concrete: microstructure, properties and materials", The Mc GrawHill Companies.
9. Jayant D. Bapat (2013), Mineral admixtures in cement and concrete, Taylor and Francis group.
10. Concrete mix proportioning as per IS 10262:2009 – Comparison with IS 10262:1982 and ACI 211.1-91 M.C. Nataraja and Lelin Das
11. IS10262-1982 Recommended Guidelines for Concrete Mix Design, Bureau of Indian Standards, New Delhi, 1998.
12. IS456-2000 Plain and Reinforced Concrete- Code of Practice, Bureau of Indian Standards, New Delhi, 2000.

RCE 013 Geoenvironmental Engineering (L-T-P 3-1-0) Credit - 4

UNIT-1 Fundamentals of Geoenvironmental Engineering

Scope of geoenvironmental engineering - multiphase behaviour of soil – role of soil in geoenvironmental applications – importance of soil physics, soil chemistry, hydrogeology, biological process – sources and type of ground contamination – impact of ground contamination on geoenvironment - case histories on geoenvironmental problems. **[8]**

UNIT-2 Soil-Water-Contaminant Interaction

Soil mineralogy characterization and its significance in determining soil behaviour – soil-water interaction and concepts of double layer, forces of interaction between soil particles, concepts of

unsaturated soil, water flow in saturated and unsaturated zone, soil-water-contaminant interactions and its implications, factors effecting retention and transport of contaminants. **[8]**

UNIT-3 Waste Containment System

Evolution of waste containment facilities and disposal practices, Site selection based on environmental impact assessment ,different role of soil in waste containment, different components of waste containment system and its stability issues , property evaluation for checking soil suitability for waste containment . **[8]**

UNIT-4 Contaminant Site Remediation

Site characterization, risk assessment of contaminated site, remediation methods for soil and groundwater, selection and planning of remediation methods. **[8]**

UNIT-5 Advanced Soil Characterization

Contaminant analysis, water content and permeability measurements, electrical and thermal property evaluation, use of GPR for site evaluation, introduction to geotechnical centrifuge modeling. **[8]**

Text

1. Yong, R. N., "Geoenvironmental Engineering, Contaminated Soils, Pollutant Fate, and Mitigation" CRC Press, New York, 2001.
2. Sharma H.D. and Reddy K.R., "Geoenvironmental Engineering: Site Remediation, Waste Containment, and Emerging Waste Management Technologies" John Wiley & Sons, Inc., USA, 2004.
3. Fredlund D.G. and Rahardjo, H., "Soil Mechanics for Unsaturated Soils" Wiley-Interscience, USA, 1993.
4. Mitchell, J. K., "Fundamentals of Soil Behaviour" Wiley, 2005.
5. Hillel D., "Introduction to Environmental Soil Physics" Academic Press, New York, 2003.
6. Rowe R.K., "Geotechnical and Geoenvironmental Engineering Handbook" Kluwer Academic Publications, London, 2000.
7. OP Gupta, " Environmental chemistry" Khanna Publishing House
8. Reddi L.N. and Inyang, H. I., "Geoenvironmental Engineering, Principles and Applications" Marcel Dekker Inc. New York, 2000.

References

1. Hillel D., "Introduction to Soil Physics" Academic Press, New York, 1982.
2. Sparks, D.L., "Environmental Soil Chemistry" Academic Press, New York, 2002.
3. O.P. Gupta, Environmental Chemistry, Khanna Publishing House

4. Bagchi, A., "Design of landfills and integrated solid waste management" John Wiley & Sons, Inc., USA, 2004.
5. Alvarez-Benedi J. and Munoz-Carpena, R., "Soil-Water-Solute Process Characterization: An Integrated Approach" CRC Press, New York, 2005.
6. Berkowitz, B. Dror, I. and Yaron, B., "Contaminant Geochemistry" Springer, Germany, 2008.
7. Mohamed, A. M. O., "Principles and Applications of Time Domain Electrometry in Geoenvironmental Engineering" Taylor and Francis, New York, 2006.

REV 551: Air Pollution Control Engineering Lab (L-T-P 0-0-2) Credit -1

1. Stack monitoring.
2. Ambient air monitoring for PM_{10}, SO_2 and NO_2.
3. Measurement of meteorological parameters (wind velocity, wind direction, humidity, temperature, solar in solution, rain fall) and drawing wind rose diagram.
4. Determination of HC and CO.
5. Determination of O_3.
6. Laboratory scale study on few air pollution control devices.

References:

1. W.H.O.: Selected Methods of Measuring Air Pollutants.
2. Pandey and Carney: Environmental Engineering
3. Keshav Kant and Rajni Kant, "Air Pollution and Control Engineering"Kannna Publishing House.

REV 552: Solid Waste Management Lab (L-T-P 0-0-2) Credit -1

1. Sample preparation and sampling techniques.
2. Coning and quartering method.
3. Profile sampling of municipal solid waste.
4. Analysis of solid waste/sludge for moisture content.
5. Analysis of solid waste /sludge for particle size.
6. Analysis of solid waste/sludge for calorific value
7. Determination of Ninthe sample.

REV 553: Environmental Sanitation Lab (L-T-P 0-0-2) Credit -1

1. Study of Storm Water Collection practices in your city.
2. Study of Sewage Collection practices in your city.
3. Study of rain water harvesting practices in your city.
4. Study on Constructions of Sewers, Joints, Manhole and other Sewer Appurtenances by site visit.
5. Layout of Sewer Lines using AUTO CAD, Planning of Sewerage System.

6. Study of sanitation in rural areas: visit to nearby village.

REV554: Building Planning & drawing Lab (L-T-P 0-0-2) Credit -1

Drawing and drafting of following with CAD software

1. Introduction to the tools and commands of drafting software.
2. Working in layers, blocks, x-ref, drawing layout and print setup. 3.3D drafting and rendering
4. Planning and drafting of elevation and cross section of door and window
5. Planning and drafting of plan and cross section of Doglegged and open well staircase.
6. Planning and Drawings of Residential buildingof1 room set (plan and section).
7. Planning and drawing of 3 room residential building with staircase.
8. Preparation of details general arrangement drawing of 4 room duplex house including planning and drafting.

SIXTH SEMESTER

ENVIRONMENTAL ENGINEERING SESSON 2018-19

Sl No.	Subject Code	Subject Name	Teaching Deptt.	L-T-P	Th/Lab Marks	Sessional		Total	Credit
					ESE	CT	TA		
1	RAS601	INDUSTRIAL MANAGEMENT	Applied Science	3—0---0	70	20	10	100	3
2	RUC601/ RAS602	CYBER SECURITY/SOCIOLOGY	Applied Science	3—0---0	70	20	10	100	3
3	REV601	DESIGN OF WASTE WATER ENGINEERING SYSTEM	Core Deptt.	3—0---0	70	20	10	100	3
4	REV602	ENVIRONMENTAL IMPACT ASSESSMENT, AUDIT AND LAWS	Core Deptt.	3—1---0	70	20	10	100	4
5	REV603	INDUSTRIAL POLLUTION CONTROL AND MANAGEMENT	Core Deptt.	3—0---0	70	20	10	100	3
6	REV061 RCE-062 REV-063	DEPTT. ELECTIVE COURSE-II HSE FOR OFFSHORE AND PETROLEUM ENGINEERS INTEGRATED WASTE MANAGEMENT FOR SMART CITIES TRANSPORT PROCESS: HEAT AND MASS TRANSFER	Core Deptt.	3—1---0	70	20	10	100	4
7	REV651	WASTE WATER ENGINEERING LAB	Core Deptt.	0—0---2	50		50	100	1
8	REV652	CAD LAB-II	Core Deptt.	0—0---2	50		50	100	1
9	REV653	QUANTITY ESTIMATION AND EVALUATION LAB	Core Deptt.	0—0---2	50		50	100	1
10	REV654	ECO-CAMP	Core Deptt.	0—0---2	50		50	100	1
	TOTAL				620	120	260	**1000**	**24**

REV 601: Design of Wastewater Engineering Systems (L-T-P 3-0-0) Credit -3

Unit 1

Introduction: Wastewater flow and its characteristics, Wastewater collection systems, Estimation and variation of wastewater flows. Problems of industrial wastewaters, Sampling protocol, Equalization, Neutralization, Proportioning processes, Volume and strength reduction. Preliminary, primary, secondary and tertiary wastewater treatment processes. Theory and design of screens, grit chambers, sedimentation, coagulation, flocculation. **[8]**

Unit 2

Physico-chemical and biological treatment strategies and their evaluation, Theory of activated sludge process (ASP), extended aeration systems, trickling filters (TF), aerated lagoons, stabilization ponds, oxidation ditches, sequential batch reactor, rotating biological contactor, etc., Mass balancing in ASP and TF and their design. **[8]**

Unit 3

Anaerobic treatment process, Effects of pH, temperature and other parameters on anaerobic treatment, Concept of anaerobic contact process, anaerobic filter, anaerobic fixed film reactor, fluidized bed and expanded bed reactors and upflow anaerobic sludge blanket (UASB) reactor. **[8]**

Unit 4

Indian standards for disposal of treated wastewaters on land and in natural streams, Agricultural irrigation, Ground water recharge, Treated wastewater reclamation and reuse, Introduction to duckweed pond, vermiculture and root zone technology for wastewater treatment, Special treatments, Recent technologies of treatment. **[8]**

Unit 5

Study on wastewater generation points, wastewater characteristics, process flow sheets, treatment scheme for tannery, sugar, textile, steel, distillery, paper/ pulp and oil refinery industry wastewater. **[8]**

Reference Books

1. Metcalf & Eddy “Wastewater Engineering: Treatment & Reuse”, Tata Mc Graw Hill.
2. Peavy, Rowe &Tchobanoglous “Environmental Engineering”, Mc. Graw Hill, New Delhi.
3. Davis, M. “Water and Wastewater Engineering’’, Mc-Graw Hill.
4. Fair, G.M. & Geyer, J.C. “Water supply and Wastewater Disposal”, John Wiley & Sons.

5. Qasim, S.R., Motley, E.M., and Zhu, G. “Water Works Engineering”, Prentice Hall Publication.
6. CPHEEO “Manual on Sewerage & Sewage Treatment”, Ministry of urban development, Government of India, New Delhi.
7. Parker, H. “Wastewater System Engineering”, Prentice Hall.
8. O P Gupta, Element of water pollution control engineering" Khanna Publishing House
9. Keshav Kant, Air Pollution Control Engineering, Khanna Publishing House
10. Rao, M.N. & Dutta, A.K. “Wastewater Treatment”, Oxford & IBH Publishing.
11. M. P. Ponia and SC Sharma, " Environmental Engineering" Khanna Publishing House.

REV 602: Environmental Impact Assessment and Audit (L-T-P 3-1-0) Credit -4

UNIT 1

Introduction to EIA & Audit, Environment & Industries, Input information, Plant operation, Environmental Management planning, Waste Streams impact on water bodies. **[8]**

UNIT2

Environmental Impact Assessment planning. Activities, Methodology for Environmental Impact Assessment, Role of Environmental Engineering firm, Role of Regulatory agencies & control boards, Role of the Public. **[8]**

UNIT3

Environmental Audit: Introduction, Environmental information Purpose & advantage of studies, General approach of environmental Auditing Environmental Audit, Audit programs in India, Auditing program in major polluting Industries, Reports of the Environmental audit studies .

UNIT4

Pollution prevention and control laws & acts: Constitution of India & environment, Constitution protection to Environment laws, Administrative & legislative arrangement for Environmental production, Indian Standards, Pollution control acts in India, critical appraisal, fiscal incentives for environmental protection. **[8]**

UNIT5

Guidelines of preparation of project report and its evaluation, methods of clearance from the concern authorities at various labels. **[8]**

References:

1. "O P Gupta, "Elements of Environmental pollution & Control Khanna Publisïng house.

2. Environmental impact assessment by Canter.

3. Environmental Chemistry by Stanley E. Manahan, VIth Ed. Lewis Publishers, London

4. Dying Wisdom: Rise, Fall, and potential of India's Traditional rain water harvesting systems by Anil Agarwal & Sunita Narayan, CSE Publication. New Delhi.

5. Environmental Impact Assessment (Theory and Practice) by Peter Wathern, Routledge (Taylor and Frances Group), London and New York.

REV 603: Industrial Pollution Control and Management (L-T-P 3 -0 -0) Credit -3

Unit 1

Industrial wastes and their sources: Various industrial processes, Sources and types of solid, liquid, gaseous wastes, Noise & radiation emissions. Sources of industrial water usages and various industrial processes requiring water use and required water quality. **[8]**

Unit 2

Processes responsible for deterioration in water quality, Various waste water streams, Control and removal of specific pollutants in industrial waste waters, e.g., oil and grease, bio-degradable organics, chemicals such as cyanide, fluoride, toxic organics, heavy metals, radio activity etc. Waste water reuse & recycling, Concept of zero discharge effluent. **[8]**

Unit 3

Control of gaseouse missions: Hood and ducts, Tall stacks, Particulate and gaseous pollutant control, Solid waste generation and disposal management.
Hazardous wastes: Definitions, concepts and management aspects. Noise & radiation: Generation, control and management. **[8]**

Unit 4

Recent trends in industrial waste management, Cradle to grave concept, Life cycle analysis, Clean technologies; Case studies of various industries, e.g., dairy, fertilizer, distillery, sugar, pulp and paper, iron and steel, metal plating, thermal power plants, etc. **[8]**

Unit-V

Environmental audit: Definition and concepts, Environmental audit versus accounts audit, Compliance audit, Relevant methodologies, Various pollution regulations, Introduction to ISO and ISO14000. **[8]**

REV 061 HSE for offshore and petroleum industries (L-T-P 3-1-0) Credit - 4

UNIT 1 - Safety assurance and assessment

Introduction to HSE - Safety assurance - Safety in design and operations -Organizing for safety - Hazard classification and assessment, Hazard evaluation and control - Hazop - Hazop case study - FMEA - Tutorial sheets **[8]**

UNIT 2 -Accident modeling, risk assessment & management

Dose assessment, safety regulations - Toxic releases- models and methods -Chemical risk analysis - Chemical exposure index (CEI) - Case studies in oil industries - Quantitative risk assessment - Fire and explosion models -Flammability diagrams - Exposure models - Fire and explosion: prevention methods - Event tree and fault tree analyses **[8]**

UNIT 3 - Environmental issues

Environmental impact and management - Impact of oil and gas industry in marine environment -Oil hydrocarbons in marine environment - Chemical disposal of offshore industry **[8]**

UNIT 4 - Environmental Management

Environmental management - Dispersion models and atmospheric pollution - Dispersion models continued - Hazard assessment **[8]**

UNIT 5 - Safety measures in design and operation

Safety measure in oil and gas industry - Safety methods in design and operation -Process safety management - Software used in HSE **[8]**

References:

1. National Safety Council Publication
2. Environmental Chemistry by Stanley E. Manahan, VIth Ed. Lewis Publishers, London
3. CPCB Green Book 4.www.moef.gov.in
4. O P Gupta, " Environmental Chemistry, Khanna Publishing House.

RCE 062 Integrated Waste Management For a Smart City (L-T-P 3-1-0) Credit - 4

Unit-1

Introduction: Solid Waste Management- Definition, Concept of 4Rs (reduce, reuse, recycle and recover) of waste management, Elements of a waste management system, Current Issues in Solid Waste Management, Integrated Waste Management Hierarchy: Source reduction, Recycling, Waste-to-Energy and Landfilling. Review of waste management under Swachh Bharat Mission and Smart Cities Program. **[8]**

Unit-2

Municipal Solid Waste: Waste Composition and Quantities**,** Collection, Transportation, Segregation, and Processing. **[8]**

Unit-3

Disposal of Municipal Solid Waste: Landfill, Biochemical Processes and Composting, Energy Recovery from Municipal Solid Waste. Municipal Solid Waste (MSW) Rules 2016. **[8]**

Unit-4

Construction and Demolition (C&D) Waste Management: Overview, Components; C&D Waste Management Rules 2016, Beneficial Reuse of C & D Waste Materials. **[8]**

Unit-5

Electronic Waste (E-Waste) Management – Issues and Status in India and Globally, E-Waste Management Rules 2016 and Management Challenges.

Hazardous Wastes: Definition, Classification, Risk assessment, Transportation of hazardous waste, Current Management Practices: Environmental audit, Containment, Remedial alternatives. **[8]**

Books:

1. George Tchobanoglous, Hilary Theisen and Samuel A Vigil, Integrated Solid Waste management, Tata McGraw Hill
2. Ramachandra T.V., *Management of Municipal Solid Waste*, 2009; by The Energy and Resource Institute, TERI
3. Sasikumar, K, Gopi Krishna, Sanoop, *Solid Waste Management*; 2009, PHI.

References:

1. Manual on Solid Waste Management, prepared by The Central Public Health and Environmental Engineering Organization(CPHEEO), India
2. MSW Management Rules 2016, Govt. of India, available online at CPCB website
3. Construction and Demolition Waste Management Rules, 2016, MoEF&CC
4. Electronic Waste Management Rules 2016, Govt. of India, available online at CPCB website.
5. Freeman, M. H.1988. Standard Handbook of Hazardous Waste Treatment and Disposal, McGraw-Hill Book Company, New York
6. http://swachhbharatmission.gov.in/sbmcms/index.htm
7. http://swachhbharaturban.gov.in/
8. O P Gupta, " Element of hazardous waste management, Khanna Publishing House.

REV 063 Transport Process: Heat and Mass Transfer (L-T-P 3-1-0) Credit - 4

UNIT 1

Introduction to transport phenomena, molecular transport mechanisms and general properties, transport with net convective flux, flow turbulence and boundary layer theory. **[8]**

UNIT 2

Transport in ducts, heat & mass transfer in duct flow, transport in immersed bodies, **[8]**

UNIT 3
Unsteady state transport and agitation. **[8]**

UNIT 4
Estimation of transport coefficient, non-Newtonian fluids, rheological characteristics of materials, agitation of non-Newtonian fluids. **[8]**

UNIT 5
Heat & mass transfer inside a porous catalyst: chemical reaction and phenomenon **[8]**

REFERENCES:

1. R. Byron Bird, Warren E. Steward and Edwin N. Lightfoot, Transport Phenomena by
2. Suhas V. Patankar, Dudley Brian Spadding, Heat and Mass Transfer in Boundary Layers, Morgan Grampian.

3. W.M. Kays, M.E. Crawford, Bernhard Weigand, Convective Heat and Mass Transfer, Mc Graw Hill
4. Asim K. Dutta, Biological and Bio-Environmental Heat and Mass Transfer, Marcel Dekker.
5. Robert Alber Greenkorn, David P. Kossler, Momentum, Heat and Mass Transfer fundamentals,
Marcel Dekker

REV 651: Wastewater Engg. Lab (L-T-P 0-0-2) Credit -1

1. Determination of color and odor of wastewater sample.
2. Determination of BOD of the wastewater sample.
3. Determination of TDS and TSS in wastewater sample.
4. Determination of COD of wastewater sample.
5. Determination of Chloride content of wastewater sample.
6. Sampling protocol of wastewater from an Industry.
7. Preservation of wastewater samples for different tests i.e., DO, BOD, Metals etc.
8. To determine the nitrogen content.

Reference Books:

1. A.P.H.A. “Standard Methods for the Examination of Water and Wastewater”, American Public Health Association.
2. Sawyer, C.N., McCarty, P.L. &Parkin, G.F. “Chemistry for Environmental Engineering”, Mc-Graw Hill.
3. Mathur, R.P. “Water & Wastewater Testing”, Lab Manual, Roorkee.

REV 652: CAD Lab-II (L-T-P 0-0-2) Credit -1

1. Design of water distribution system using available software.
2. Design of sewer line/waste water drain network using available software.
3. Digitization of your city map using available software and showing pollution map.
4. Predicting concentration of air pollutant, emitted from stack, at any given location in ambient environment by using any dispersion modeling software.

REV 653: Quantity Estimation Evaluation lab (L-T-P 0-0-2) Credit -1

1. Method of Estimation: General items of works for estimates, units and Measurement, method of accounting for the deduction of openings etc.
2. Detailed estimates of a single roomed and a two roomed residential building.
3. Analysis of rates: Definition of analysis of rates, Prime cost, and work charged establishment.
4. Quantity of materials per unit of works for major Civil Engineering items. Resource planning through analysis of rates, market rates.
5. PWD scheduled and cost indices for building material and labour.
6. Valuation: Purpose of Valuation, Market Value, Book Value, Rate able Value, Capital Cost, Capilized Value, Ideal investment, Sinking fund, Depreciation, Straight Line method, sinking fund method, quantity survey method, Valuation of building, rent fixation.

References:

1. Dutta, B.N., "Estimation and Costingin Civil Engineering (Theory and Practice)", UBS Publishers Distributers Private Ltd., New Delhi.
2. Singh, Gurucharan, Singh Jagadish, "A Text book of Estimation Costing and Valuation" Standard Publishers Distributers, Delhi-110006.
3. Peurifoy, Robert L., Oberlender, GaroldD., "Estimating Construction Costs" Tata Mc Graw Hill Education Pvt. Ltd.,New Delhi.

REV 654: Environment Camp (L-T-P 0-0-2) Credit -1

The purpose of the camp is to train students in applying modern techniques/ equipment to prepare a detailed report of selected study area. The course will be run in the form of a camp for 7 working days and will involve one or more of the following components:

1. Study of the environmental problems in the study area.
2. Sampling work and analysis in the lab.
3. Field study and primary data collection.
4. Secondary data collection from agencies.
5. Statistical analysis of data, model development and estimating pollutant quantities.

6. Designing of system using software/ model/ data.
7. Preparing a map using GIS software and report writing.

Note: The students should be given a comprehensive problem at the end which requires inputs/knowledge/application from all the units of the syllabus. It may be evaluated as a part of TA.

Books

1. Metcalf &Eddy "Waste water Engineering: Treatment & Reuse", Tata Mc Graw Hill.
2. Azad, H.S. "Industrial Waste water Management Hand book", Mc-Graw Hill.
3. Arceivala, S.J. & Asolekar, S.R. "Waste water Treatment for Pollution Control and Reuse'', Mc-Graw Hill.
4. Culp, G., George, W., Williams, R. and Mark, Hughes, V.J r. "Waste water Reuse and Recycling Technology- Pollution Technology "Review- 72, Noyes Data Corporation, New Jersey.
5. Pandey, G.N. and Corney, G.C. "Environmental Engineering", Tata Mc-Graw Hill.
6. Edmund, B. Besselieve P.E. "The Treatment of Industrial Wastes", Mc-Graw Hill.
7. Nancy, J.S. "Industrial Pollution Control: Issues and Techniques", Van Nostrand Reinhold.
8. Shen, T.T. "Industrial Pollution Prevention Hand book", Springer- Verlag.
9. Environment (protection) Act- 1986, Ministry of Environment and Forest, Government of India.
10. A K Jain, Environmental Engineering, Khanna Publishing House.